

SIPLE DOME, Antarctica

WMO: 893450

Lat: 81.653S

Lon: 148.990W

Elev: 2034

StdP: 13.65

Time Zone: -10.00 (W10)

Period: 97-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-56.7	-52.4	N/A	N/A	N/A	N/A	N/A	N/A	142.9	-27.5	141.4	-27.3	11.0	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
	DB Range	DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.9	30.4	N/A	28.5	N/A	23.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	18.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 139.3	86.3	76.1	DB	-61.7	41.1	6.6	6.2	-66.4	45.5	-70.2	49.2	-73.9	52.7	-78.7	57.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.0	13.7	3.1	-9.1	-16.1	-19.3	-17.9	-23.4	-26.5	-21.7	-10.2	5.5	14.4	(6)	
	DBStd	18.11	7.06	8.84	10.64	11.77	13.50	14.35	14.10	14.73	13.95	10.13	7.94	6.37	(7)	
	HDD50	21542	1125	1314	1832	1984	2147	2037	2275	2373	2151	1866	1335	1103	(8)	
	HDD65	27017	1590	1734	2297	2434	2612	2487	2740	2838	2601	2331	1785	1568	(9)	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	18.0	15.4	15.8	16.7	23.6	17.7	19.2	20.8	20.4	21.5	16.4	14.9	13.9	(14)	
	PrecAvg	9.6	0.6	0.5	0.7	0.8	1.0	1.1	1.1	1.1	1.0	0.8	0.5	0.5	(15)	
(16) Precipitation	PrecMax	14.7	1.3	1.1	2.7	2.9	2.4	2.4	3.8	2.6	2.8	1.3	2.0	1.7	(16)	
	PrecMin	3.7	0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1	(17)	
	PrecStd	2.8	0.4	0.3	0.6	0.6	0.6	0.6	0.9	0.6	0.7	0.4	0.4	0.4	(18)	
	PrecStd	2.8	0.4	0.3	0.6	0.6	0.6	0.6	0.9	0.6	0.7	0.4	0.4	0.4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	28.2	21.6	15.4	16.6	15.3	10.4	7.4	15.4	21.1	28.9	36.5	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	30.6	22.2	15.4	9.3	11.6	9.4	5.8	1.7	9.6	13.2	23.1	30.7	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	28.9	19.2	10.1	5.4	4.2	7.2	1.2	-0.7	4.2	8.1	20.8	28.9	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	23.3	15.9	6.2	1.1	0.3	2.1	-3.8	-5.5	-0.8	4.8	16.9	23.4	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	13.9	14.8	15.3	15.4	15.7	15.7	15.5	15.2	16.9	17.5	14.8	13.9	(35)	
		MCDBR	19.2	19.4	16.0	17.9	18.5	17.1	18.5	16.2	18.3	20.9	19.6	20.2	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.215	0.193	0.146	N/A	N/A	N/A	N/A	N/A	0.138	0.180	0.199	0.212	(40)	
	taud		2.219	2.218	2.066	N/A	N/A	N/A	N/A	N/A	2.019	2.203	2.243	2.234	(41)	
	Ebn at Noon		305	281	218	0	0	0	0	0	216	288	313	316	(42)	
	Edh at Noon		29	21	11	0	0	0	0	0	11	22	28	31	(43)	
(44) All-Sky Solar Radiation	RadAvg		2606	1491	364	0	0	0	0	0	158	1055	2425	3038	(44)	
	RadStd		70	47	16	0	0	0	0	0	8	30	81	80	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page